

Content additional attachments

- 1) Belt tensioning instruction (PI-10-25-00-02)
- 2) Labelling and lifting (PI-00-10-00-02)
- 3) Torque applied, bolts & screws (BPS-20-35-10-02)
- 4) Pulley alignment using TKBA40 laser alignment tool (PI-10-25-00-31)
- 5) Removal and fitting of hpt seals (PI-10-25-00-13)

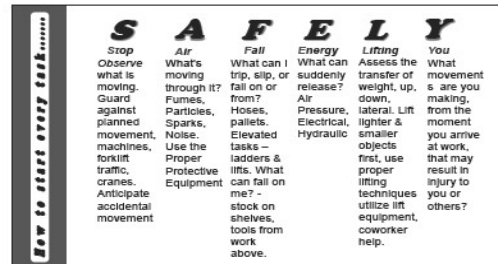
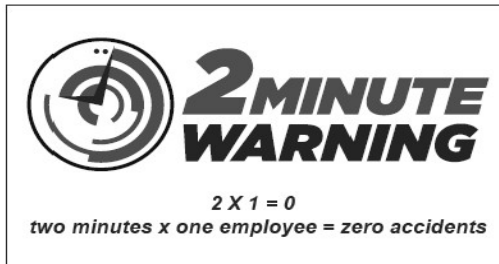
V - BELT TENSIONING

PROCEDURE

ATTENTION POINTS



BEFORE CONTINUING, PLEASE ENSURE YOU HAVE READ AND UNDERSTOOD THE RELEVANT S.O.P.



To carry out the adjustment and tensioning of the belts the following tools are required.

- ❖ 1 x Torque wrench up to a range 210Nm
- ❖ 1 x 24mm socket.
- ❖ 2 x 30mm open ended spanners
- ❖ 1 x 24mm spanner.
- ❖ 2 x 10mm spanners
- ❖ 1 x belt tensioning indicator Bearward part No BE6400011100 – This indicator is our recommended item and what this instruction is based on. To purchase contact +44 (0) 1604 762851 / BEN_Spares@wabtec.com. If an alternative method is used, the force and deflection parameters still apply. If the belt is tensioned to a value outside the parameters then **the belt warranty will be void.**

Before any work is carried out make sure the emergency stop button has been pressed and the battery cables removed.

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V - BELT TENSIONING

PROCEDURE

1. To access the pulley belts and fasteners on the idler system, remove the pulley guarding using 10mm spanners.

2. Slacken off the four M16 fasteners on the idler system using the 24mm torque wrench and spanner.

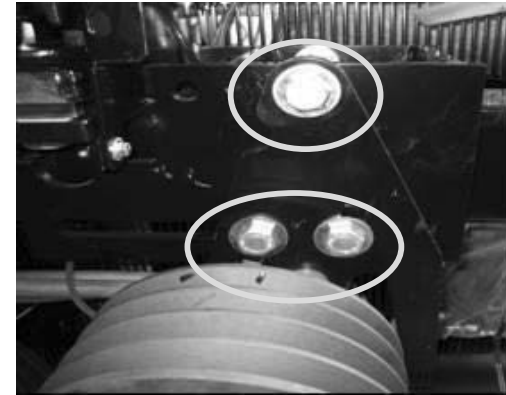
NOTE: - Prior to tensioning, check the tensioning device (threaded rod OR turnbuckle tensioner) for lubricant. The threaded portion on both should be lubricated in the working range with ROCOL RTD Compound.

NOTE: - For the correct tension, follow the instructions on the General Assembly drawing for the specific radiator. The deflection distance and force will be listed. If the force is given in Newtons (N) then convert to kgf by dividing by 9.81.

3. Using the belt tensioning indicator set the lower marker ring at the deflection distance required in mm on the lower scale

4. Set the upper marker ring against the bottom edge of the top tube.

ATTENTION POINTS



NOTE: 4th bolt is behind the top bolt in this picture.



Bearward part No BE6400011100
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V - BELT TENSIONING

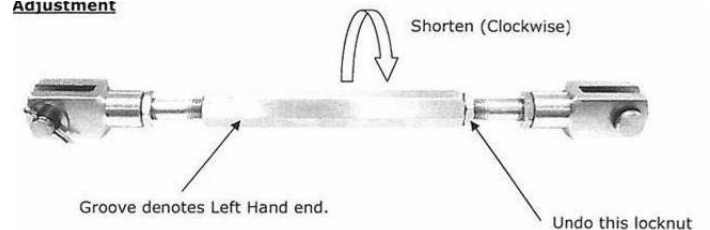
PROCEDURE

6. Place the belt tension indicator on top of the belt at the centre of span, and apply a force at right angles to the belt deflecting it to the point where the marker ring is level with the top of the adjacent belt.
7. **ADJUSTING USING THE TURNBUCKLE ARRANGEMENT:**
 - 7.1. Adjust the turnbuckle to alter the tension on the belt.
 - 7.2. To adjust belt tension; using an M30 open ended spanner undo the locknut on turnbuckle.
 - 7.3. Using the M30 spanner (with the left hand thread to your left) turn the hexagon bar. Adjust clockwise to shorten, counter-clockwise to lengthen.
 - 7.4. Adjust the hexagon turnbuckle to the desired length / belt tension value. (Refer to installation drawing.)

ATTENTION POINTS



Adjustment



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PI-10-25-00-02-BeltTensioning

V - BELT TENSIONING

PROCEDURE

- 7.5.** Read of the force value indicated by the top edge of the upper marker ring.
- 7.6.** Compare this force to the kgf value shown on the drawing.
 - 7.6.1.** If the value is correct to the drawing reset the locknut on turnbuckle and re-torque all fasteners on the idler to 210Nm
 - 7.6.2.** If the value is below the recommended value follow steps **7.1** through **7.6**.

ATTENTION POINTS



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PI-10-25-00-02-BeltTensioning

V - BELT TENSIONING

PROCEDURE

8. ADJUSTING USING THREADED ROD TENSIONER

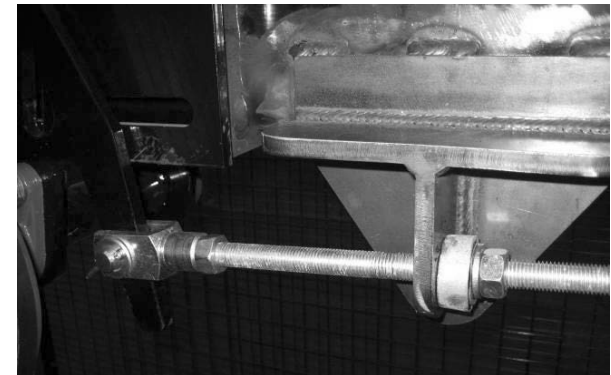
- 8.1.** To adjust belt tension; using a 24mm AF combination spanner adjust the locknut on the threaded rod.
- 8.2.** Use the 24mm spanner to turn the nut. Adjust clockwise to shorten the rod and increase belt tension, counter-clockwise to lengthen and decrease tension.
- 8.3.** Adjust the tensioner until the desired belt tension value is reached. (Refer to installation drawing.)

NOTE: - DO NOT OVER TENSION BELTS.

- 8.4.** Read the force value indicated by the top edge of the upper marker ring.
- 8.5.** Compare this force to the kgf value shown on the drawing.
 - 8.5.1. If the value is correct to the drawing re-torque all fasteners on the idler to 210Nm
 - 8.5.2. If the value is below the recommended value repeat steps **8.1** through **8.5**.

ATTENTION POINTS

LOCKNUT



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V - BELT TENSIONING

PROCEDURE

9. Re-fit all guarding around pulleys.
10. Run engine for 20 mins then re-check tension.
11. Re-set engine to on line function.

❖ Note:- for single belt drives or multi groove belts a straight edge should be placed across the two pulleys to act as a datum for measuring the amount of deflection.

ATTENTION POINTS



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PI-10-25-00-02-BeltTensioning



This shows how the unit should be lifted via the fitted lifting 45° brackets, once it has been stood up.

It is important to maintain a 45° slinging angle on the chains when lifting using these brackets

These instructions are for lifting/turning the Radiator only—DO NOT Attempt to lift/turn unit when attached to any other component.



GENERIC TURNING/ LIFTING INSTRUCTION

Bearward Engineering Ltd. accepts no responsibility for any consequential damage or injury caused by the execution of this instruction

Bearward Engineering Ltd. Can provide technicians to advise/train personnel in line with all of our procedures.

For further information please contact us at:
BEN_Drawing_Office@wabtec.com

Before attempting to lift/turn any unit—check lifting labels for lifting eye/turning locations

Check Accessories for Lifting Brackets/Lugs



This label tells you that this lifting point is to be used for a straight lift and for turning the radiator. These points should be fitted with an M30 4 ton collar eye bolt

When a lifting hole/bracket is fitted with this label the unit must only be lifted in the direction shown i.e. lifting the unit in a flat condition off the lorry.



RADIATOR
LIFT ONLY

BL15/35.14

Lifting holes should be fitted with M24 2.5 ton collar eye bolts .

If 4 labels are fitted then all 4 points should be used at the same time—**NEVER** use just 2.

Ensure slinging angle of chains is at 45 degrees.



DO NOT USE
LIFTING POINT

BL0098-5822

These lifting points **MUST NOT** be used.

They were for manufacture of unit, and not designed to lift whole unit with.